Posets (Partially Ordered Set):

A relation on a set S is called a partial order if it is

(1) reflexive ↓ $aRa \ \forall a \in S$

(2) antisymmetric ↓ $aRb \ \& \ bRa \Rightarrow a = b$

(3) transitive ↓ $aRb \ \& \ bRc \Rightarrow aRc$

- Partial order is denoted by $\propto$
- $(S, \propto) \rightarrow$ partially ordered set or POSET
- $a \propto b \rightarrow$ a preceeds b or b suceeds a

Ex– $(Z, \leq) \rightarrow$ POSET

(1) $a \leq a \ \forall \ a \in S$ ∴ Reflexive

(2) $a \leq b \ \& \ b \leq a \Rightarrow a = b$ ∴ Antisymmetric

(3) $a \leq b \ \& \ b \leq c \Rightarrow a \leq c$ ∴ Transitive

Ex- $(P(S), \subseteq) \rightarrow$ POSET?
$\downarrow$
power set of S

$S = \{a, b\}$; $P(S) = \{ \{\}, \{a\}, \{b\}, \{a, b\} \}$

$\forall s \in P(S) \Rightarrow s \subseteq s$
If $S_1 \subseteq S_2$ & $S_2 \subseteq S_1 \Rightarrow S_1 = S_2$
If $S_1 \subseteq S_2$ & $S_2 \subseteq S_3 \Rightarrow S_1 \subseteq S_3$

$\therefore$ POSET

Ex- $(\mathbb{Z}, /)$ $m/n \rightarrow$ m divides n
Is this a POSET?
$\Rightarrow$ No. as $a/(-a)$ and $(-a)/a$ but $a \neq -a$
i.e; not antisymmetric.

Note: $(\mathbb{Z}^+, /) \rightarrow$ POSET?

$a/a \rightarrow k_1$ $\forall a \in \mathbb{Z} \rightarrow$ Reflexive
a/b & b/a only if $a = b \rightarrow$ Antisymmetric

a/b & b/c $\Rightarrow$ $b = k_1 a \Rightarrow c = k_2 b$
$\Rightarrow c = k_1 k_2 a$
$\Rightarrow \frac{a}{c}$

Note: If a given function is POSET, then we can draw its Hass Diagram.

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Comparability: If $a \preceq b$ and or $b \preceq a$ then a & b are comparable element of $(S, \preceq)$ else they are not comparable.

e.g. $(Z^+, |)$ $2|4$ $\therefore$ 2 & 4 are comparable
$2 \nmid 5$ $\therefore$ 2 & 5 are not comparable

In $a \preceq b$

a → predecessor of b
b → successor of a

In $a \preceq c \preceq b$

$a \rightarrow$ predecessor of b, c & immediate predecessor of c

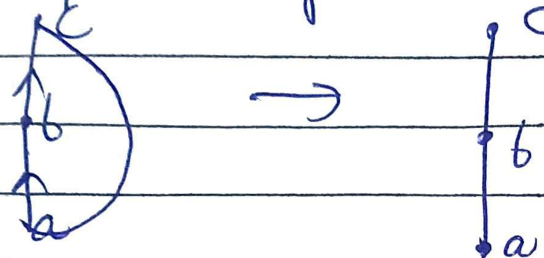
Totally Ordered Set (Linearly Ordered Set): If every two elements of S are comparable, then poset $(S, \preceq)$ is called a totally ordered set. This is also called a chain.

E.g. $(Z, \leq)$ is a totally ordered set or chain.

Hasse Diagram: A poset $(S, \preceq)$ can be represented by means of a diagram called Hasse Diagram.
Here, if $x \preceq y$ then y is placed at a higher level than x and joined by a straight line.
Steps:
1) Start with a directed graph of relation.
2) Remove the loop at all vertices.
3) If aRb, then b appears above a and is

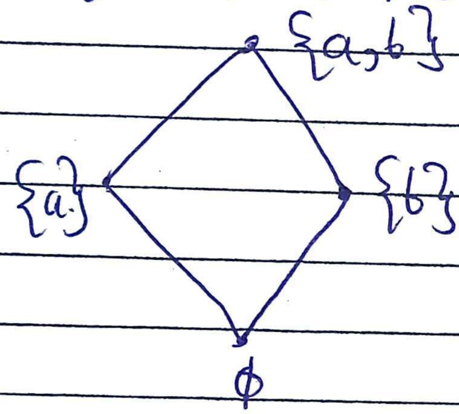
connected by an edge with arrow upwards.

4) Remove the arrows & edges whose existence is implied by transitive property.

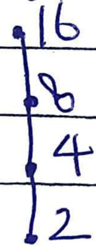


Ex- $A = \{a, b\}$

$P(A) = \{\phi, \{a\}, \{b\}, \{a,b\}\}$ $(P(A), \subseteq)$



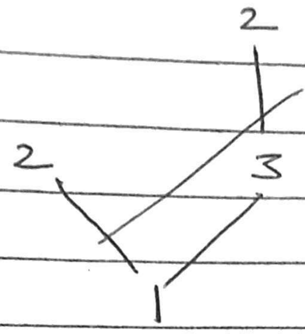
Ex- Let $A = \{2, 4, 8, 16\}$. Then $(A, |)$ is a poset.

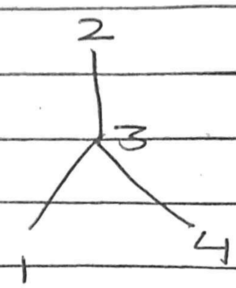


11/03/26 Ex-2) $A = \{1, 2, 3, 4\}$; $R = \{(1,1), (1,2), (1,3), (2,2), (3,2), (3,3), (4,2), (4,3), (4,4)\}$

Prove that R is partial order and draw Hasse Diagram.

$\Rightarrow$ (A, R) is a POSET.

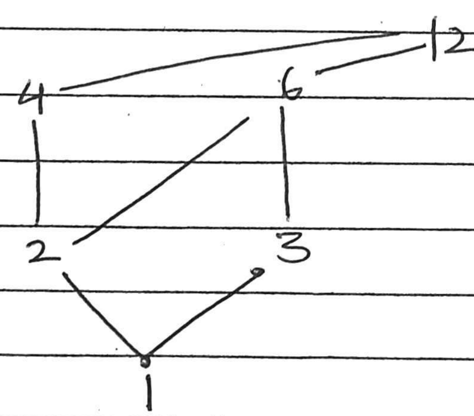


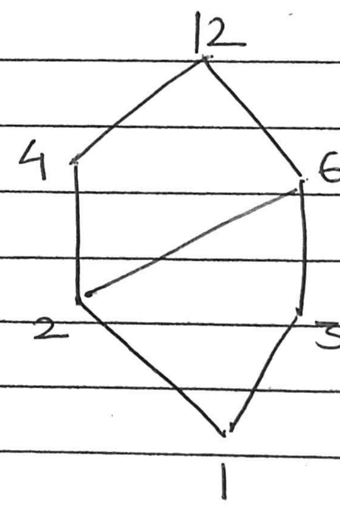


Q3) $(D_{12}, /)$

↳ Set of all divisors of 12

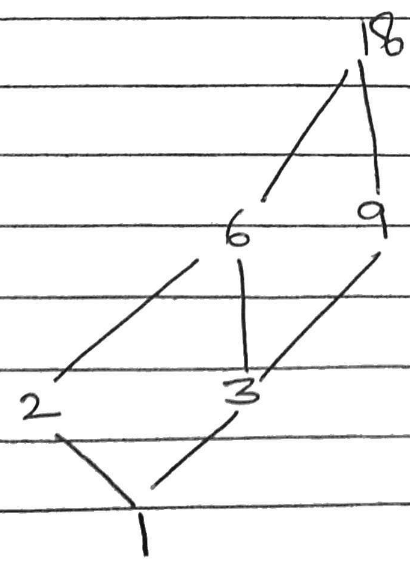
$D_{12} = \{1, 2, 3, 4, 6, 12\}$



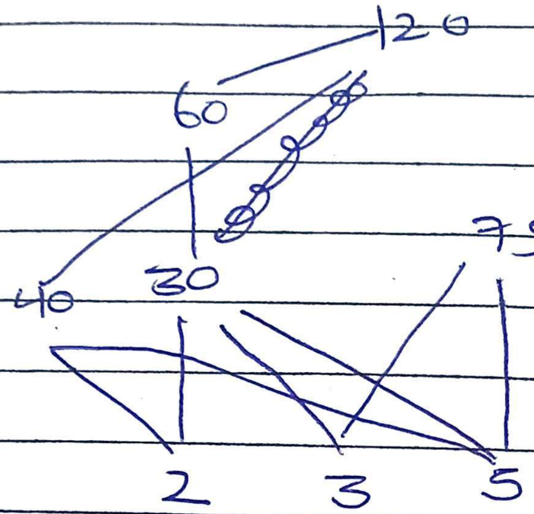


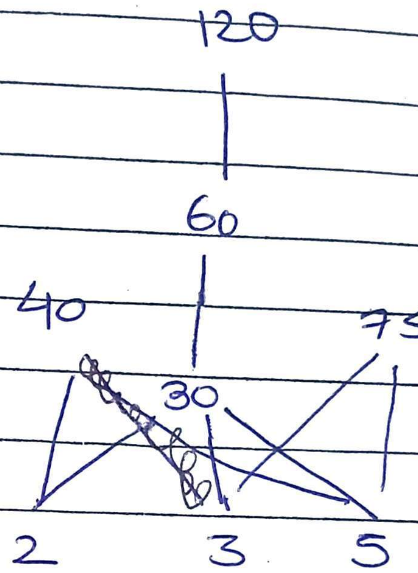
Q4) $(D_{18}, /)$

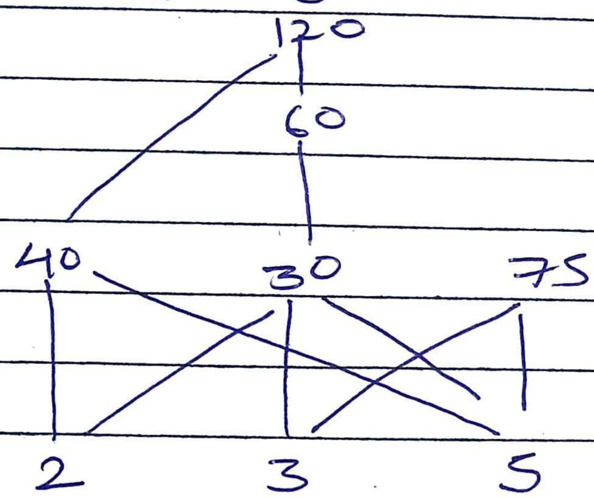
$D_{18} = \{1, 2, 3, 6, 9, 18\}$



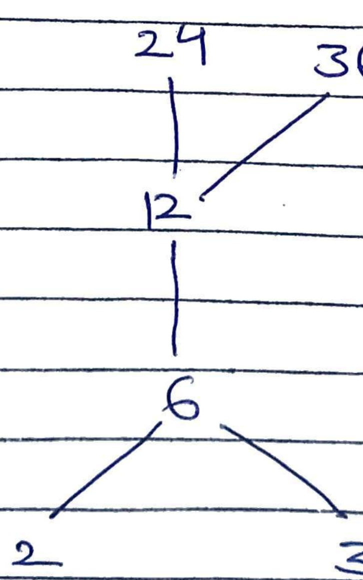
Q.5) Let $S = \{2, 3, 5, 30, 40, 60, 75, 120\}$
$(S, /)$

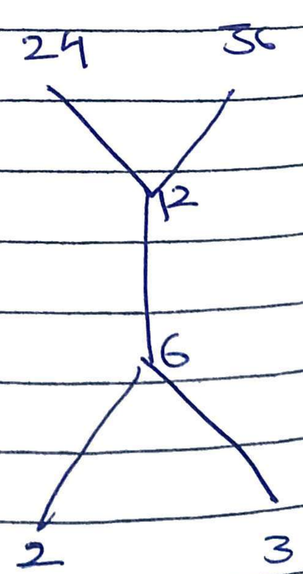






Q. $A = \{2, 3, 6, 12, 24, 36\}$; $(A, /)$





Special Elements in POSET

Let $(S, \preceq)$ be a poset

Maximal element: $a \in S$ is maximal element if $\nexists\, x \in S$ such that $a \prec x$

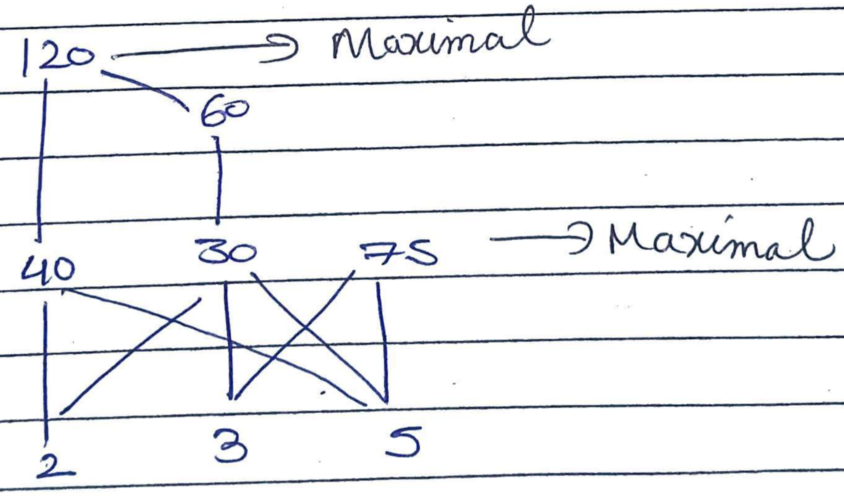
Minimal element: $b \in S$ is minimal element if $\nexists\, x \in S$ such that $x \prec b$

Greatest element: $a \in S$ is greatest element of x if $x \preceq a \;\forall x \in S$

Least element: $b \in S$ is greatest element if $b \preceq x \;\forall x \in S$.

If greatest element exists, it'll be unique and similarly for least element.

Ex-



Greates & least DNE in this diagram.

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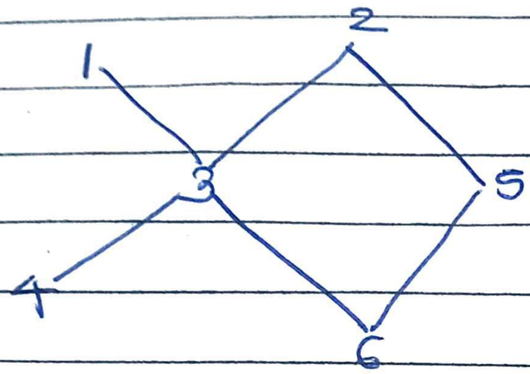
Q. $A = \{1, 2, 3, 4, 5, 6\}$



Find : a) All maximal and minimal element of A.
b) Greatest and least element of A.
c) All linearly ordered subsets of A each of which contains atleast 3 elements.

a) Maximal : 1, 2 Minimal : 4, 6

b) Greatest & Least : DNE

c) $\{4, 3, 1\}$, $\{4, 3, 2\}$, $\{6, 3, 2\}$, $\{6, 3, 1\}$, $\{6, 5, 2\}$

Q. Consider the POSET : $\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \subseteq\}$.

Find a) maximal element b) minimal element c) All upper bounds of the set and least upper bound if it exists. d) All lower bounds of $\{1, 3, 4\}$ and greatest lower bound if it exists.

$\Rightarrow$ a) maximal element : $\{2, 3, 4\}$, $\{1, 3, 4\}$, $\{1, 2\}$
b) minimal element : $\{1\}$, $\{2\}$, $\{4\}$

c) $\{2, 4\}$, $\{2, 3, 4\}$
Least upper bound $- \{2, 4\}$

d) {1,4}, {3,4}, {4}, {1}

Greatest lower bound DNE